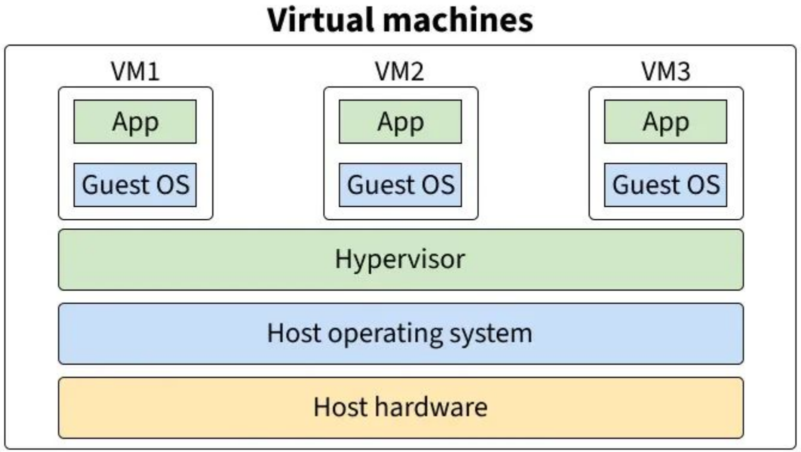
# Virtual Machine

**Definition:** A virtual machine (VM) is a digital version of a physical machine that functions as an isolated system with its own virtualized hardware resources and operating system.

## VM vs Local Machine:

| Local | VM (EC2) |
|---|---|
| Runs on your laptop | Runs on AWS data center |
| Only you can access | Has public IP, accessible worldwide |
| Shuts down when you close it | Can run 24/7 |
| Uses your hardware | Uses AWS hardware |



# SSH & Security Groups

**SSH (Secure Shell):**

- Encrypted connection to remote servers
- Uses key pair authentication (.pem file)
- Command: ssh -i cs6650-key.pem ec2-user@<IP>

**Security Groups:**

- Acts as virtual firewall for EC2
- Controls inbound/outbound traffic
- e.g. I opened: Port 22 (SSH) and Port 8080 (HTTP)

# Elastic IP & Instance Types

**Elastic IP:**

- A static, fixed public IP address
- Your EC2 IP stays the same after restart
- Costs money if NOT attached to a running instance

**Instance Type (t2.micro):**

| Spec | Value |
|---|---|
| vCPU | 1 |
| Memory | 1GB |
| Network | Low to Moderate |

**Why it matters:** Bigger workloads need bigger instances (and cost more).

# GCP & AWS

| Aspects | GCP (Cloud Run) | AWS (EC2) |
| --- | --- | --- |
| Setup | Near zero config | Manual VM, Security Group, SSH |
| Scaling | Automatic | Manual |
| Billing | Per request | Per hour |
| Port/SSH | Managed for you | Configure yourself |

**Takeaway:**
- GCP: Easier, more abstracted
- AWS EC2: More control, more complex

# Cross Compile & SCP & Testing

**Cross Compilation:**
GOOS=linux GOARCH=amd64 go build -o newmain main.go

- `GOOS=linux` → Target OS is Linux (EC2 runs Linux)
- `GOARCH=amd64` → Target architecture is 64-bit
- Compiles on Mac, runs on EC2

**Test:**

```
scp -i cs6650-key.pem newmain ec2-user@<IP>:~        # Upload to EC2
./newmain                                            # Run on EC2
curl -v http://<your public IP address>:8080/albums  # Test from local
```

**Why 0.0.0.0?** It allows external requests, not just localhost